

Lesson 1: Charge

Conservation, sign, and polarization

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

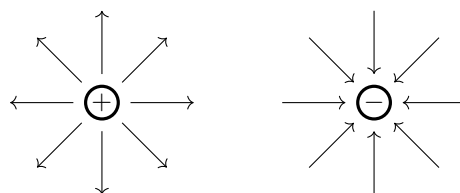
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Transparent tape
- Balloon or plastic comb
- Wool cloth
- Empty aluminum can

Predict first

Will two tape strips prepared the same way attract or repel? What result would distinguish two signs of charge?



Arrow direction is the force on a positive test charge.

Investigate

1. Prepare two equal tape strips the same way; lift and suspend them side by side. Record attraction or repulsion.
2. Stack two fresh strips, lift together, then separate. Compare top-to-top, bottom-to-bottom, and top-to-bottom.
3. Charge a comb; approach a neutral can without touching and map the distance at which it begins to roll.
4. Repeat three times, changing only distance.

Explain from the evidence

Like charge signs repel and unlike signs attract. Rubbing and peeling redistribute charge rather than creating net charge. A charged object can attract a neutral object because charges in the neutral material shift slightly; the nearer opposite response can dominate. Humidity can drain charge, so a weak result is evidence to record, not permission to invent a result.

Change one thing

Compare dry and humid conditions, or two insulating materials. Keep geometry and preparation count fixed.

Evidence record

Prediction

One change

Observation

Claim + because

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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